Modular-coordinate guards and a corrected early formula

2026-10-27 | T-5518 | Observation report; no formal hypothesis decision.

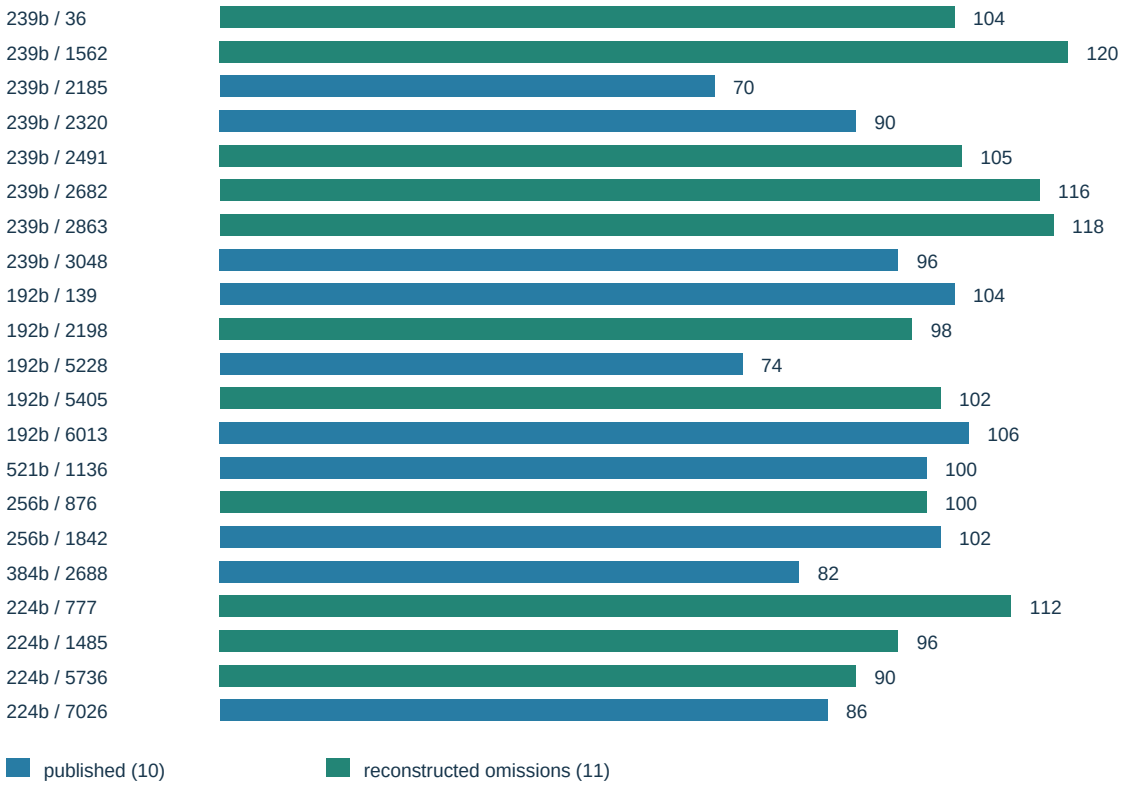
The historical selector remains unknown. All 11 reconstructed omissions and all 10 publications pass the tested coordinate/fiber guards. Literal use of a recovered draft error fails every curve, including all publications.

1. Fixed cohort, expanded coordinate coverage

21 immutable curve identities; six recovered streams. Available later LiDIA source tables:61 odd-prime levels 3 through 293, 46 canonical and 15 Atkin. The source is not authenticated1997 code. Full identities and field/model/order/seed data are in comparison.csv.

Independently verified: 1281 first-stage cells, 2568 Bezout identities including controls; every cohort derivative gcd is constant and every F(0,j) is nonzero. FLINT independently proves completeness of 1185 rational parameters. All 2071 source/Fricke-target/Atkin fibers have constant checked derivative gcds and contain neither j0 nor j1728.

Recorded fibers checked per curve; every checked exception count is zero



CANONICAL-ATKIN-ALL-FIBERS-01. Unit: fibers; n=21 curves, 61 levels. Exact counts; no error bars.

Two constructed ordinary curves over F_1009 produce three first-stage exceptional guards and three exceptional fibers among five tested control fibers. Source-reader controls bind all 61 tables. The negative cohort result is therefore accompanied by actual detectable exceptions.

2. The CCR coefficient correction, tested on all 21 curves

Three author-hosted URLs supplied the same 24-page Charlap-Coley-Robbins draft. Morain 1995, printed page 275, explicitly corrects its formulas. The recovered scan version is uncertain: papers cite the draft as 1991/1992, while scan metadata is 2018. Pages 8-9 and the correction were visually compared after OCR.

Both texts define p_k as the sum over one representative from each pair of nonzero kernel points: p_k=sum x(iP)^k, i=1..(ell-1)/2. Thus p_0=(ell-1)/2. This convention is aligned before testing.

Coefficient	Literal draft, page9	Derived / Morain correction
Anew-A	5*(p2+A*p0)	-10*(3*p2+A*p0)
Bnew-B	7*(5*p3+3*A*p1+2*B*p0)	-14*(5*p3+3*A*p1+2*B*p0)

The corrected coefficients follow directly from the same draft's equations 4.4, 4.7 and 5.1. Frozen CCR-LITERAL-COEFFICIENTS-01 evaluates degree 13 on all 21 curves and one constructed ordinary F_101 control.

Measured / independently verified: 22 full degree 84 division-polynomial algebras;132 multiple-coordinate relations;44 corrected k1/k2 identities; exact Bezout common-zero certificates for both literal residuals. Every common-zero gcd has degree 0.

Derived coverage: every one of the 14 cyclic degree 13 subgroups over the algebraic closure per curve, 294 for the cohort. This follows from the full division-polynomial certificate; individual subgroup enumeration and full point counts were not performed.

Observed label result: the literal formula is inconsistent for 11/11 omissions and 10/10 publications. A universal rejection rule based on it cannot preserve the published set. A conditional caller, different degree, root-pair policy or other implementation remains outside this test.

Evidence-flow diagram only; its arrows assert no isogeny map or historical program use.

3. Custody, replay and remaining scope

Raw archive: 148 sealed files; 56, 570, 199 bytes. Protocols, all inputs, native outputs, source-table bytes and the GPL notice, controls and failed attempts are retained.

Archive SHA-256:
6a1d693b88e7c1eb61f6c61e3203b98a972b6e160b3640bdc75db82568f66629

The exact cohort digest and full UID/model/order/seed rows are retained in BUNDLE.json and comparison.csv.

Retained failures and amendments

Three preparation attempts misread table framing and stopped before native curve calls. Earlier verifier attempts assumed monic gcds and suffixed trace paths; exact output and source semantics required unit-scaled gcds and pre-suffix paths. Those attempts are preserved. UID binding was strengthened without changing outputs. One derived CSV join was repaired using curve_uid; no native producer reran and the sealed archive did not change.

Native wall times are retained in report.md as diagnostic modern ARM64 observations under uncontrolled load. They are not historical timing or performance comparisons. Operational failure does not count as mathematical rejection.

Portable verification

After extracting evidence.tar.zst:
`python3 -B evidence/verify.py`
For root completeness, use python-flint 0.9.0 and add `--fibers`. The first audit is standard-library only; the full fiber audit uses in-process FLINT. Neither launches an external producer or contacts the network.

Local LMFDB manifests were complete. The genus 1, q=1009 query returned no rows; controls were counted directly and no catalog crosswalk is claimed. The av_fq extraction ended 2026-10-19T16:45:49.243703+00:00; cryptographic fields are outside the selected index.

Limits and next discriminating evidence

This study tests coordinate singularities and one literal printed error. It does not execute a complete level 293 point counter, recover the original caller, or establish kernel/eigenvalue/CRT stopping policy and candidate visitation. The P-224 cube/splitting observation is still distinct from proof of historical use. A dated caller/configuration or an independently predictive rule is still required.

Object-store delivery was refused by the canonical HeadObject403 guard. Git delivery uses the existing documented large-file-ok exception. No remote store success or green merge gate is claimed. The PR has inherited immutable A/A-record collisions, recorded separately.

Primary sources: Morain 1995, JTNB7, 255-282, doi:10.5802/jtnb.143; CCR draft in the author archive; pinned LiDIA source. Full URLs, exact source digests and public-export exclusions are in report.md and SOURCE-BOUNDARY.json.